AI Phishing Detection Dashboard

Date: 15 July 2026
Author/Lead: Ankit Bansal (B.Tech CSE - AI & ML)
Status: Completed Prototype
Repository: https://github.com/Ankit-Bansal-coder/AI-Phishing-detection

1. Executive Overview

Objective: Develop an AI-powered dashboard that helps users identify phishing URLs before visiting suspicious websites.

Challenge: Phishing websites are becoming increasingly difficult to identify, putting users at risk of credential theft and financial fraud.

Solution: The dashboard uses Machine Learning models (Logistic Regression and Decision Tree) to classify URLs as legitimate or phishing. It also provides explainable AI insights, stores scan history using SQLite, and includes a sandbox simulation for cybersecurity awareness.

Business Impact: The project serves as an educational cybersecurity tool and demonstrates practical integration of web development and Machine Learning.

2. Technical Architecture & Technology Stack

Frontend: React.js, Vite, HTML, CSS, JavaScript
Machine Learning: Python, Scikit-learn, Logistic Regression, Decision Tree
Database: SQLite
Workflow: User URL → Feature Extraction → ML Prediction → Risk Score & Explanation → Scan History → Dashboard.

3. Key Deliverables & Features

- AI-based phishing URL detection.
- URL feature extraction engine.
- Explainable AI visualization.
- SQLite scan history.
- Interactive model playground.
- Sandbox simulation.
- Responsive dashboard interface.

4. Implementation Highlights & Performance

The project follows a modular architecture with separate frontend and ML components.

Security: Educational sandbox, no real phishing execution.
Performance: Lightweight React application with fast predictions.
Quality: Modular codebase, reusable components, documented setup.

5. Challenges Faced

- Selecting meaningful URL features.
- Integrating Python-trained models with React.
- Designing explainable prediction results.

• Managing scan history using SQLite.

6. Future Scope

• Train using larger real-world datasets.
• Add Random Forest/XGBoost.
• Integrate Google Safe Browsing or VirusTotal APIs.
• Deploy on Vercel.
• Add user authentication and cloud storage.

7. Setup Instructions

1. Clone the repository.
2. Run **npm install**.
3. Run **npm run dev**.
4. (Optional) Install Python dependencies using **pip install -r python/requirements.txt**.
5. Execute **python/train.py** to retrain the models.

8. References

• React Documentation
• Vite Documentation
• Scikit-learn Documentation
• SQLite Documentation
• OWASP Phishing Guidance

9. Conclusion

This project demonstrates a practical implementation of Machine Learning for phishing detection using React, Python, Scikit-learn, and SQLite. It strengthened my understanding of cybersecurity, web development, and AI integration while providing a useful educational prototype.